THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS (OHONGSS)
KIGOMA REGION

FORM FOUR MOCK EXAMINATION

041 **BASIC MATHEMATICS**
TIME: 3 HOURS **YEAR: 2024**

Instructions

1. This paper consists of sections A and B with a total of **fourteen (14)** questions.
2. Answer **all** questions in both section A and section B.
3. Each question in section A carries **six** (**6**) marks while each question in section B carries **ten** (**10**) marks.
4. **All** necessary working and answers for each question done must be shown clearly.
5. NECTA Mathematical tables may be used where necessary.
6. Non programmable calculators may be used.
7. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
8. Write your **examination number** on every page of your answer sheet(s).

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SECTION A (60 Marks)

Answer **all** questions in this section.

1. (a) At a disco party club a red light flashes on every 56 minutes and a green light flashes on every 70 minutes. If the two light flashes are switched on at the same time;
   (i) After how long will both light flashes on together?
   (ii) If the party was on Sunday 07:56 morning when will they flash again?

   (b) Amina decided to go out for shopping. She used $\frac{1}{4}$ of her savings to buy chocolate, $\frac{2}{3}$ of the remaining to buy ice cream. If she was left with 4200/=. Find amount of money she had at the beginning of the shopping?

2. (a) Calculate the value of x in the equation $25(2logx) = x$

   (b) Rationalize the denominator $\frac{1}{3\sqrt{2}-2\sqrt{3}}$.

3. (a) Given that $\varepsilon = \{x: -3 \leq x < 9\}$, $A = \{x: -1 \leq x < 5\}$ and $B = \{x: x \text{ is a whole number less than or equal to nine}\}$ Find the number of subset of $(A^c \cap B)$

   (b) The interior angle of a regular polygon is one hundred and eight degrees greater than the exterior angle. How many sides does the polygon have?

4. (a) if the line 4y-3x-18=0 is passing through the point (k,6), find the value of k.

   (b) Find the value of x in the following equation $2^{x+5} = \left(\frac{1}{2}\right) \times 32^{-x}$

5. If triangle ABC is similar to PQR and their sides are $\overline{AB} = 5cm, \overline{BC} = 4cm$, $\overline{AC} = 6cm$ and $\overline{PR} = 18cm$.

   (a) Draw the triangles and find the length of the other sides of a similar triangle

   (b) Prove that triangle ABC is similar to PQR

6. (a) The number of the square tiles needed to surface the floor of a hall varies inversely as the square of the length of a side of the tile used. If 2016 tiles of sides 0.4 m would be needed to surface the floor the hall. How many tiles of sides 0.3 m would be required?

   (b) The price of Radio in USA is US dollar 650 and in London the same Radio costs Euro 450. If the exchange rate is one Euro is Equivalent to 2.45 US dollars, work out the difference between the cost of radio in USA and in Landon. Give your answer in Euros.

7. (a) A damaged table that costs Tsh. 20,000 was sold at a loss of 15% .Find the selling price.

   (b) The following information is related to Johson a trader as at 12/12/2022.

   Opening stock……...$Sh.$ 90,000

   Cost of Sales …… 75 % of sales

   Closing stock……...20% of cost of goods sold.

   Net profit ……..…. 20% of sales.

   Sales ……...340,000/=

   Calculate:-

   (i) Purchases (ii) Cost of Sales (iii) Closing stock (iv) Expenses

8. (a) The seventh term of Arithmetic Progression (A.P) is 27 and the tenth term of the same Arithmetic Progression (A.P) is 60.Find the first term and common difference.

(b) Beatrice deposited Tsh 2,000,000 in a bank at the rate of 8% compounded annually for 2 years.

(i) Find how much she receives as compound interest at the end of the second year of her investment.

(ii) Calculate the interest after two years.

9. (a) Mwanahawa is a students at a certain school at Liwale District she is walking up a five hundred meters high of a building . The base of a building has an inclined plane of thirty degrees. How far she will walk to reach to the top of the building (your answer in kilometers)

(b) If $\frac{a}{b} = \sqrt{3}$ find the value of $\frac{a^2-b^2}{a^2+b^2}$.

10. The entrance door of a certain class at Liwale day Secondary school has two stairs, the length of the base stair is twelve plus a certain number while its height is two plus the same number. However, the second stair's height is four times the same number from the first stair, while its length is four plus the same number. Moreover, the second stair is on top of the base stair

(a) By using the concept of mathematics draw a well diagram to represent the Information above.
(b)Write the expression for the area of the down stair and the upper stair.
(c) What is the value of the number if the total area of the stairs is $104cm^2$

SECTION B (40 Marks)

Answer **all** questions in this section

11. (a)The score of 40 candidates in mathematics examination were as follows .

86, 56, 19, 31, 36, 32, 61, 67, 37, 34, 20, 28, 43, 27, 76,

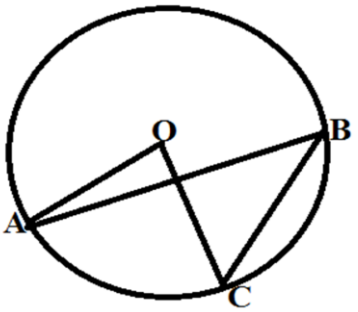
09, 85, 06, 68, 38, 18, 27, 41 , 71 ,27, 53 , 48 , 11, 23, 69

68, 54, 34, 56, 75, 45, 20, 35, 40, 41

(i) Use the class intervals of $1-10$, $11-20$ etc to prepare the frequency distribution table

(ii) Use the assumed mean of the class mark of the class interval of $31-40$. to calculate the mean scored.

(b) Carefully study the figure below and then find the value of angle AOC reflex
$O\ is\ the\ center\ of\ a\ circle\ \ and\ \angle ABC = 65^0$



12.(a) Show that the radius of a circle with the length of an arc of πcm and the central angle of $\frac{\pi}{6}$ is 6cm

(b) An aero plane take off from Dar es Salaam $(60^0S, 30^0E)$ to Dodoma $(50^0S, 30E)$ at $10knots$.Determine the time used to reach Dodoma.

(c) A ship sail start a journey from$(40^0N, 28^0E)$ and sails to 1000 nm due East. Find the its new Latitude and Longitude

13. (a) Solve the following system of linear equation by using inverse matrix method.

$3x + 2y - 4 = 0$

$2x = 5 + y$

(b) A translation T takes the point (0, 6) to (0, 10). Find where it will take the point (2, -3) about the origin.

(c)Use the appropriate diagram to find the coordinates of the final image when the point P $(2, -3)$ is first reflected along the line $y = x$ followed by another reflection along x axis.

14. Elizabeth and Agnes are trailors. They make x blouses and y skits each week. Elizabeth does all the cutting and Agnes does all the sewing. To make a blouse it takes 5 hours of cutting and 4 hours of sewing. To make a skirt it takes 6 hours of cutting and 10 hours of sewing. Neither tailor works for more than 60 hours a week.

(a) For sewing show that $2x + 5y \leq 30$

(b) Write down another inequality in x and y for the cutting

(c) If they make at least 8 blouses each week, write down another inequality.

(d) Using 1cm to represent 1 unit on each axis, show the information in part (a), (b) and (c) graphically. Shade only the required region.

(e) If the profit on blouse is Tsh. 3,000/= and on a skirt is Tsh. 10,000/=, calculate the maximum that Elizabeth and Agness can make.